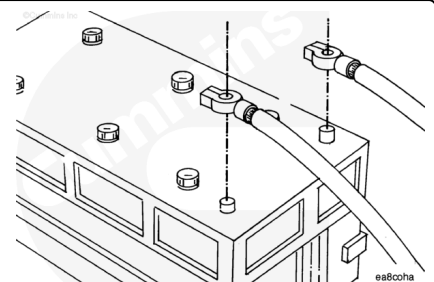


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Disconnect the batteries or air supply line to the air starter to prevent accidental engine starting.

Drain the coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

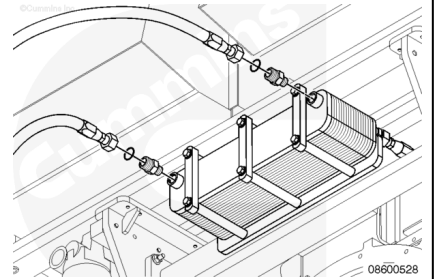
Drain the oil from the marine gear oil cooler. Refer to the marine gear manufacturers instructions for draining the oil.

Remove

Disconnect marine gear oil lines from the marine gear oil cooler.

Remove and discard the o-rings.

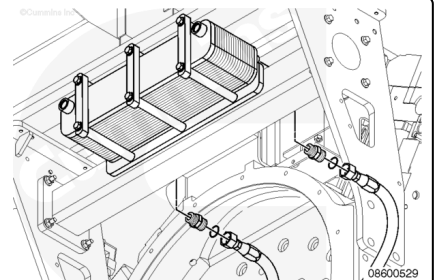
Remove the oil line connectors from the marine gear oil cooler.



Disconnect the coolant lines from the rear of the marine gear oil cooler.

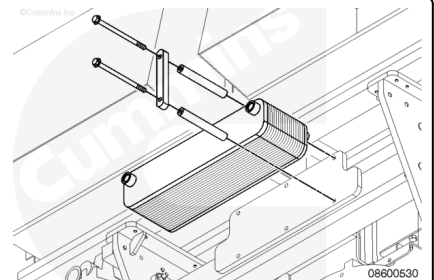
Remove and discard the o-rings.

Remove the coolant line connectors from the marine gear oil cooler.



Remove the six capscrews, three front brackets and spacers that hold the marine gear oil cooler to the rear bracket.

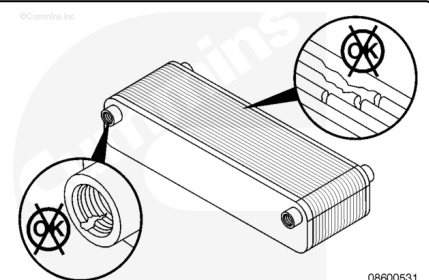
Remove the marine gear oil cooler.



Inspect for Reuse

Inspect the marine gear oil cooler plates and shell for damage. If the plates or shell is damaged the marine gear oil cooler **must** be replaced.

Inspect the marine gear oil cooler coolant and oil ports for thread damage. If the threads are damaged beyond repair the marine gear oil cooler **must** be replaced.

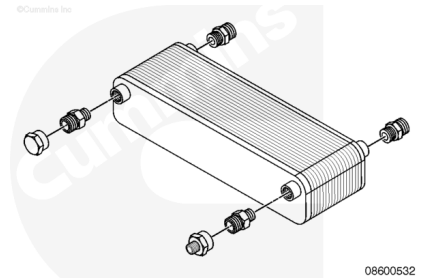


Pressure Test

Install the coolant and oil connectors into the marine gear oil cooler.

Place a Joint Industry Counsel 20 (15/8-12) threaded cap onto one of the oil connectors of the marine gear oil cooler.

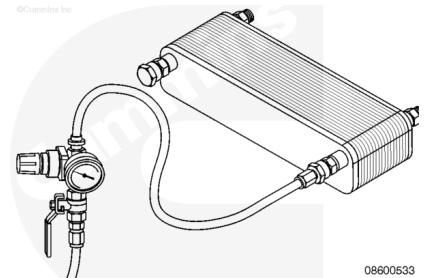
Place a Joint Industry Counsel 20 (15/8-12) threaded cap with an air fitting onto the remaining oil connector of the marine gear oil connector.



Install a pressure regulator onto the cap and air fitting.

Connect shop air to the pressure regulator.

Adjust the pressure regulator to 170 kPa [25 psi].



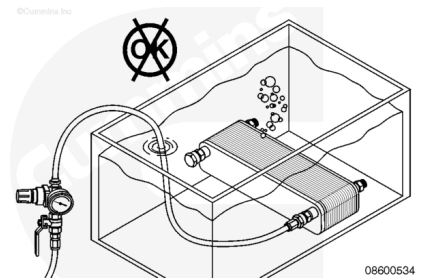
⚠ CAUTION ⚠

To reduce the possibility of damage to the marine gear oil cooler, the water in the test tank must be treated with a noncorrosive additive.

Submerge the marine gear oil cooler in a tank of treated water for 10 minutes.

If air bubbles escape from the fuel side of the marine gear oil cooler in the last two minutes of the test, the marine gear oil cooler **must** be replaced.

Remove the marine gear oil cooler from the tank of treated water.

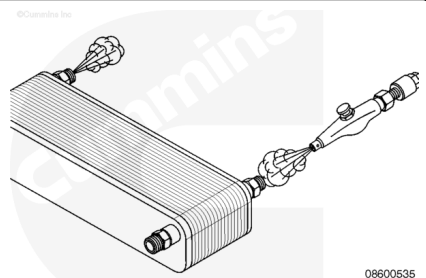


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying dirt and debris can cause personal injury.

Remove the pressure regulator from the air fitting.

Remove the caps from the fuel connectors of the marine gear oil cooler.



Drain the treated water out of the marine gear oil cooler.

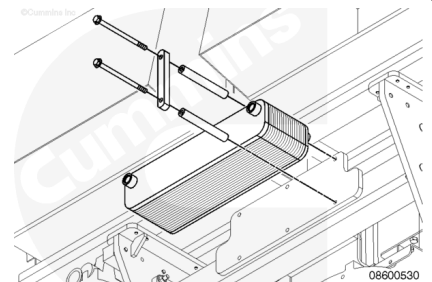
Use compressed air to dry the coolant side of the marine gear oil cooler.

Install

Place the marine gear oil cooler onto the rear bracket and install the front brackets, spacers and capscrews.

Tighten the capscrews.

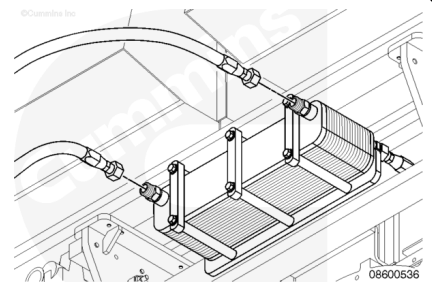
Torque Value: 45 n•m [33 ft-lb]



Connect the coolant and gear oil lines to the marine gear oil cooler connectors.

Tighten the coolant and gear oil lines.

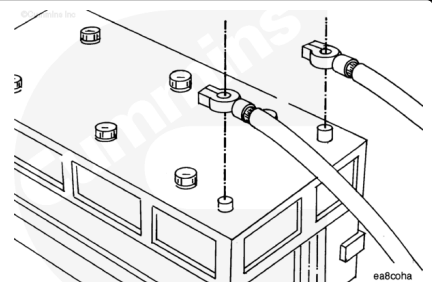
Torque Value: 74 n•m [55 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries or air starter air supply line.

Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

Add gear oil to the marine gear. Refer to the manufacturer's instructions.

Last Modified: 26-Apr-2004
